



Georgia Tech  
College of  
Engineering

# COE 3001

# MECHANICS OF DEFORMABLE BODIES

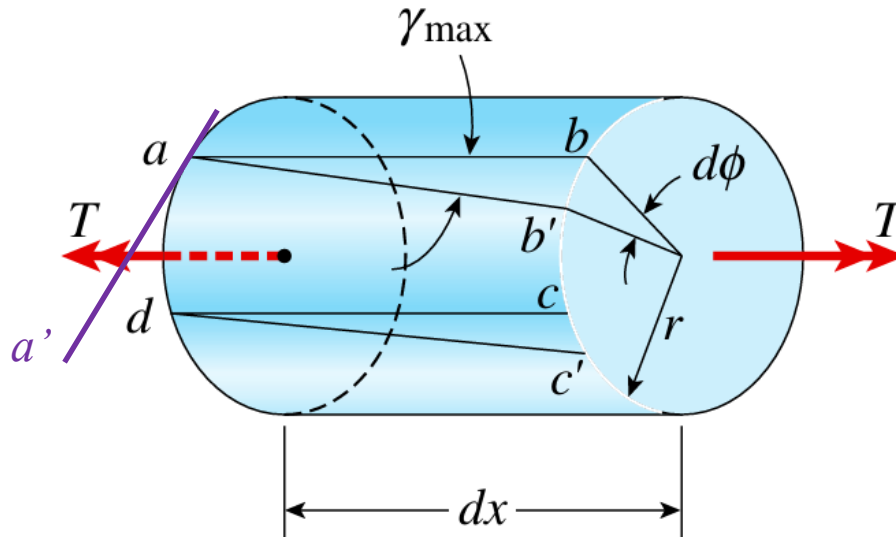
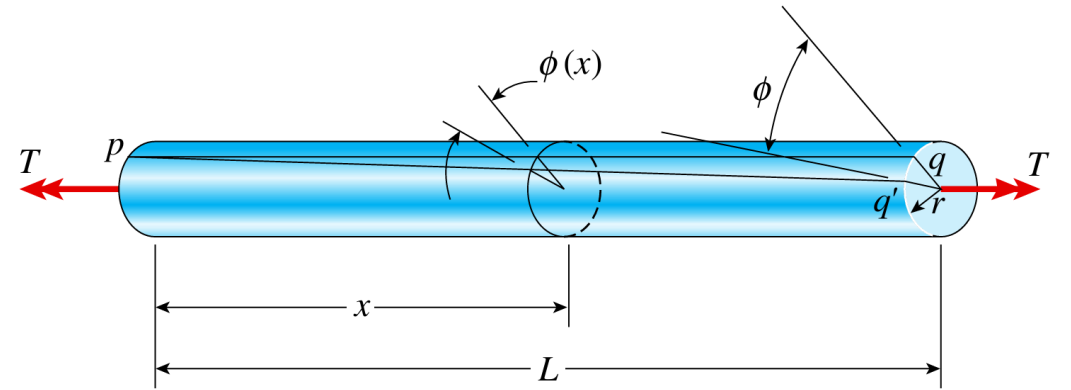
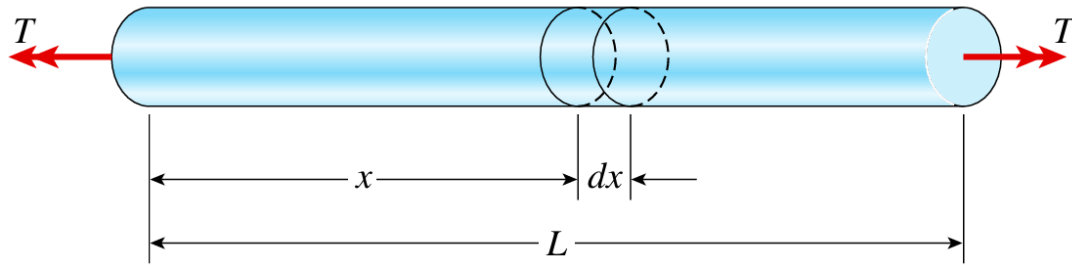
*Lecture 9* – Torsion – torsion formula cont.

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# Shear strains



$$\gamma_{\max} = \frac{bb'}{ab} = \frac{r d\phi}{dx}$$

Denote rate of twist:

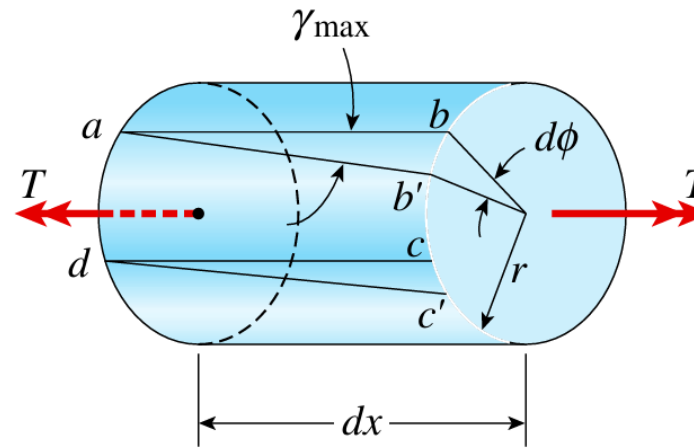
$$\theta(x) = \frac{d\phi(x)}{dx}$$

$$\Rightarrow \gamma_{\max} = r\theta$$

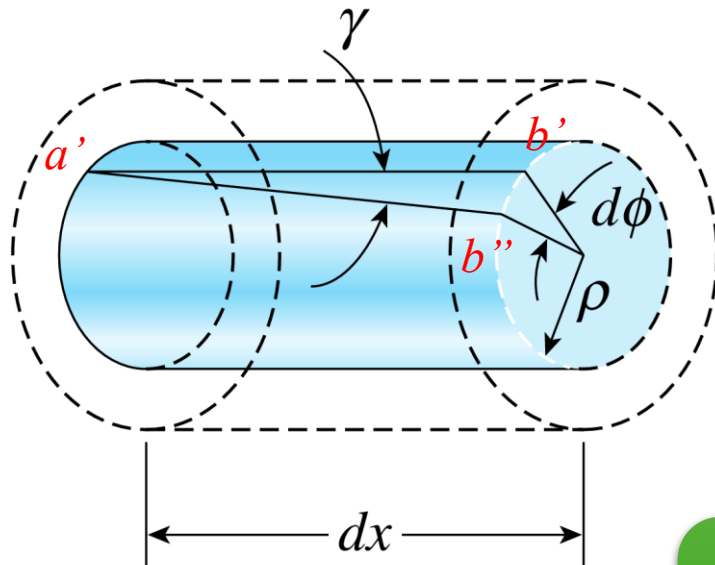
At fixed x,

$$\theta = \frac{d\phi}{dx} = \text{const.}$$

# Shear strains



$$\gamma_{\max} = \frac{bb'}{ab} = \frac{rd\phi}{dx} = r\theta$$



$$\gamma = \frac{b'b''}{a'b'} = \frac{\rho d\phi}{dx} = \rho\theta$$

$$\gamma = \frac{\rho}{r} \gamma_{\max}$$

The relation between deformation and strain

Linear distribution of shear strain

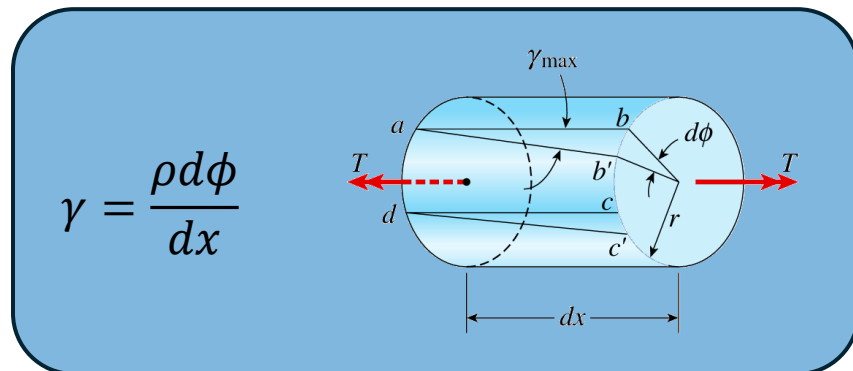
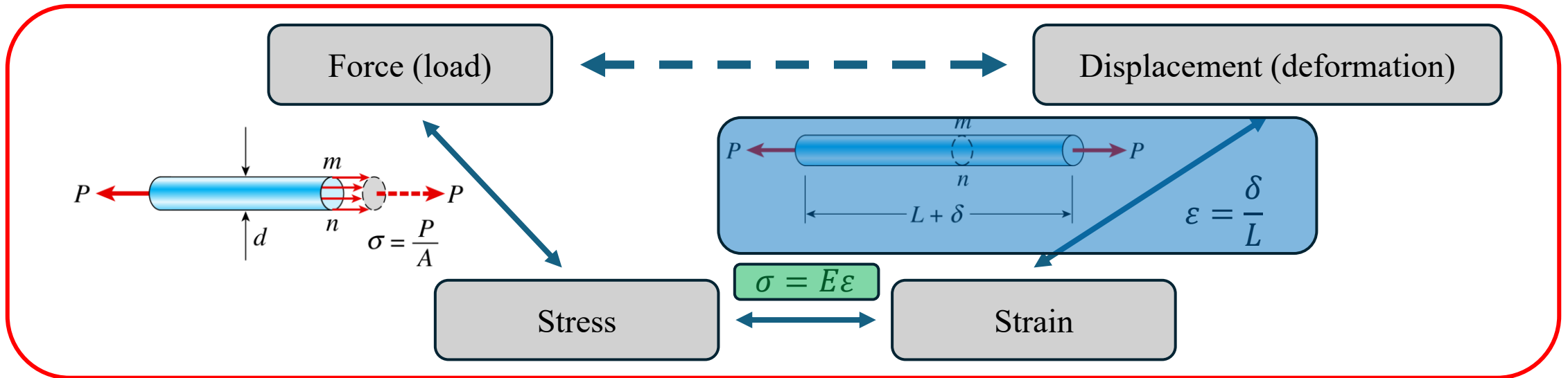
In the special case of **pure torsion**, the rate of twist is a constant

$$\theta(x) = \frac{d\phi(x)}{dx} = \frac{\phi}{L}$$

Therefore, for pure torsion only,

$$\gamma_{\max} = \frac{r\phi}{L}$$

# Torque-AoT relations



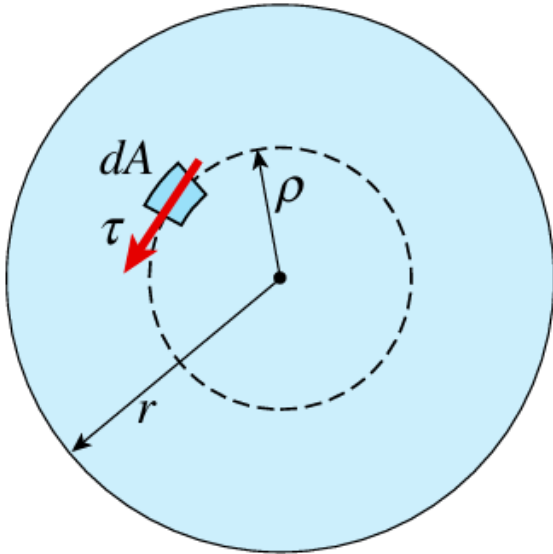
$$\tau = G\gamma$$



$$\tau = G\rho\theta = G\rho \frac{d\phi}{dx} = \frac{\rho}{r} \tau_{\max}$$

$$\tau_{\max} = Gr\theta = Gr \frac{d\phi}{dx}$$

# The torsion formula (circular shaft ONLY)



$$\tau = G\rho\theta = G\rho\frac{d\phi}{dx} = \frac{\rho}{r}\tau_{\max}$$

$$\tau_{\max} = Gr\theta = Gr\frac{d\phi}{dx}$$

$$dM = \rho \times \tau dA = \frac{\tau_{\max}}{r} \rho^2 dA$$

AoT:

$$T = \int_A dM = \frac{\tau_{\max}}{r} \int_A \rho^2 dA$$

$$\theta = \frac{d\phi}{dx} = \frac{\tau}{G\rho} = \frac{T}{GI_p}$$

Denote

Polar moment of inertia

Torsional rigidity

$$I_p = \int_A \rho^2 dA$$

Then we have

$$T = \frac{\tau_{\max}}{r} I_p$$

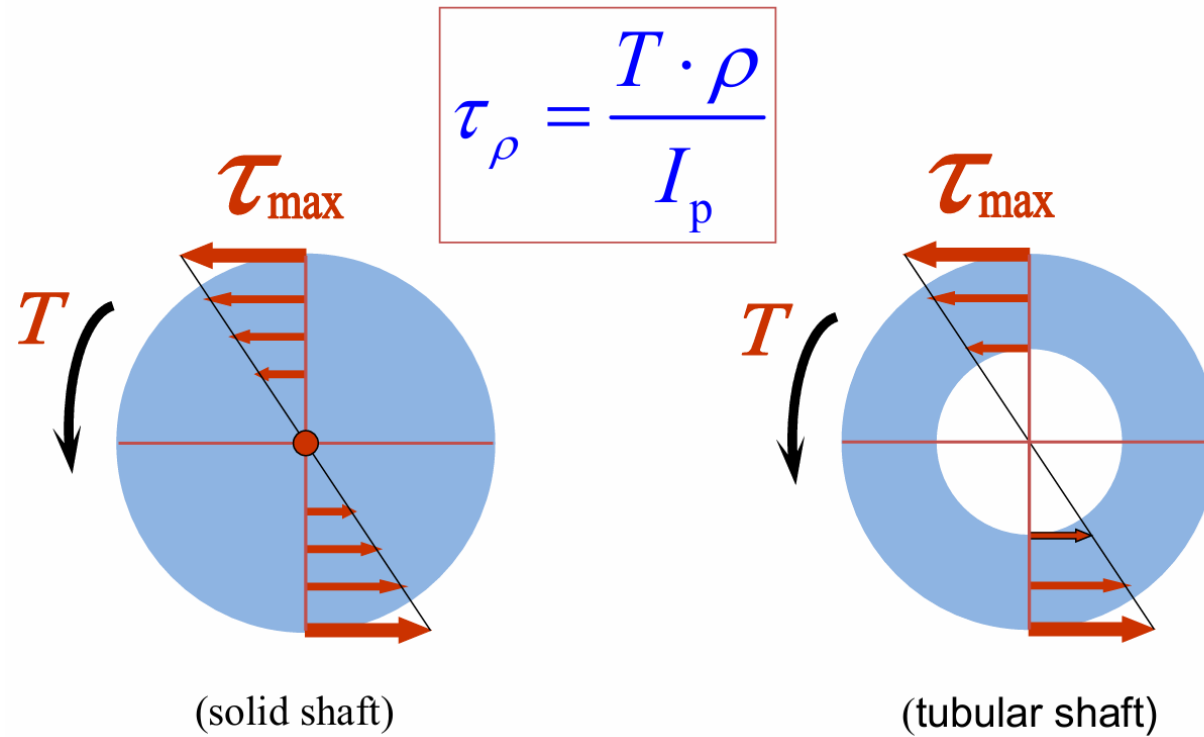
For a bar in pure torsion:

$$\phi(x) = \int_0^x \theta dx = \frac{Tx}{GI_p}$$

$$\Leftrightarrow \tau_{\max} = \frac{Tr}{I_p} \Rightarrow \tau = \frac{T\rho}{I_p}$$

$$\phi = \int_0^L \theta dx = \frac{TL}{GI_p}$$

# Shear distribution



Summary:

$$\tau = \frac{T\rho}{I_p}$$

$$\phi = \int_0^L \frac{T(x)}{GI_p(x)} dx = \frac{TL}{GI_p}$$

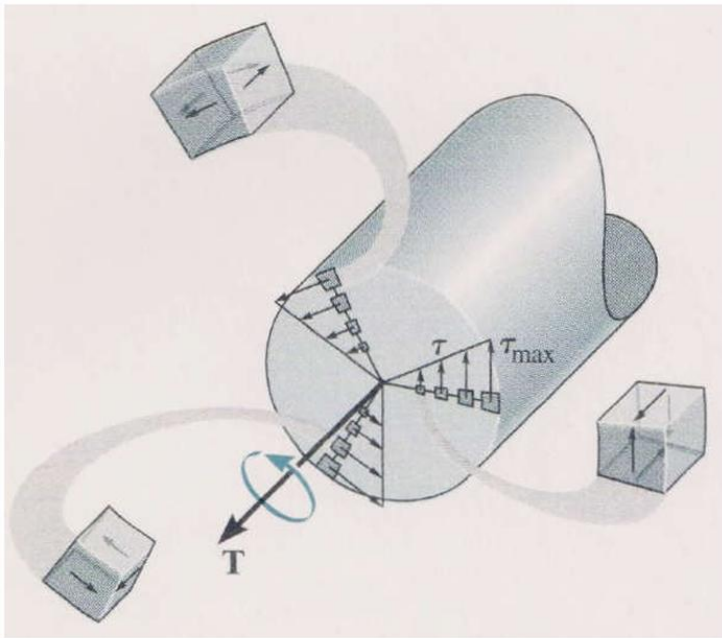
$GI_p$ : Torsional rigidity

$\frac{L}{GI_p}$ : Torsional flexibility

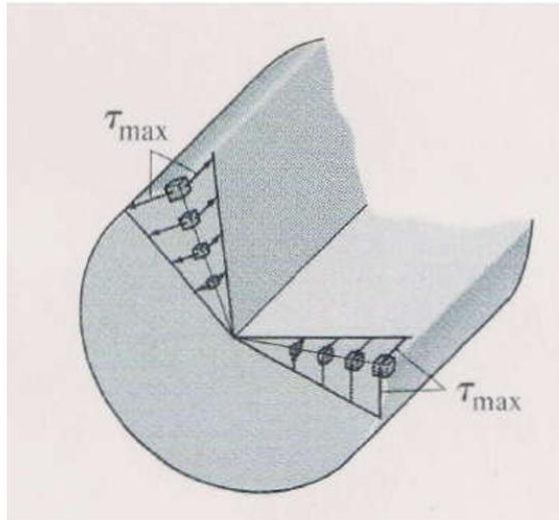
$\frac{GI_p}{L}$ : Torsional stiffness

# Shear stress along longitudinal plane

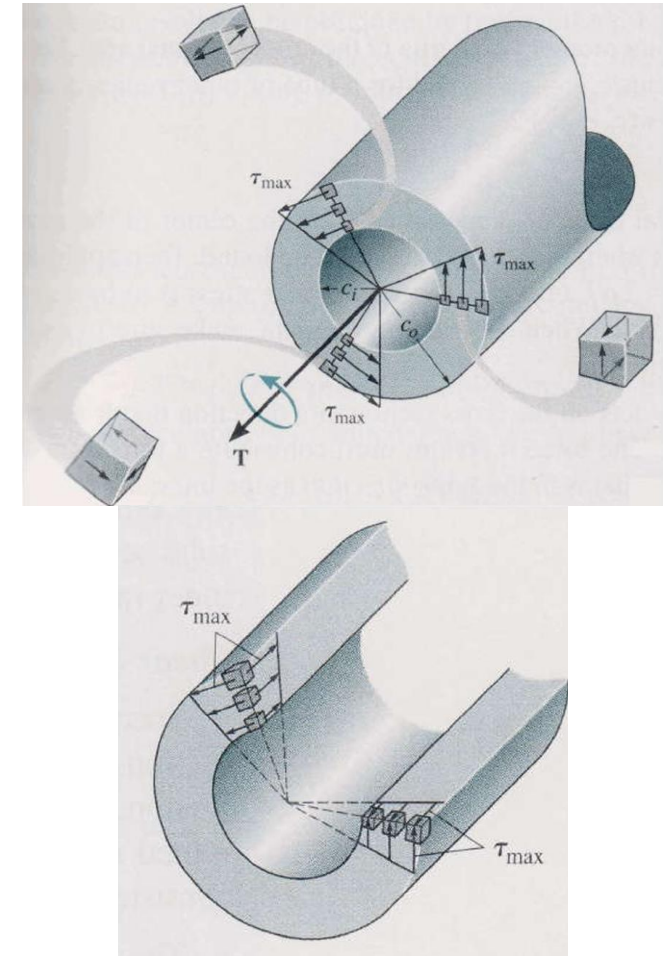
Complementary property of shear (Equality of Shear Stresses)



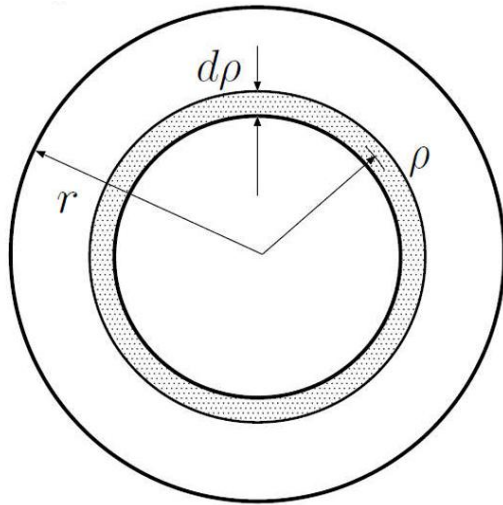
**linear distribution** of shear stress in the **cross-sectional plane**



associated shear stress distribution along an **longitudinal plane**



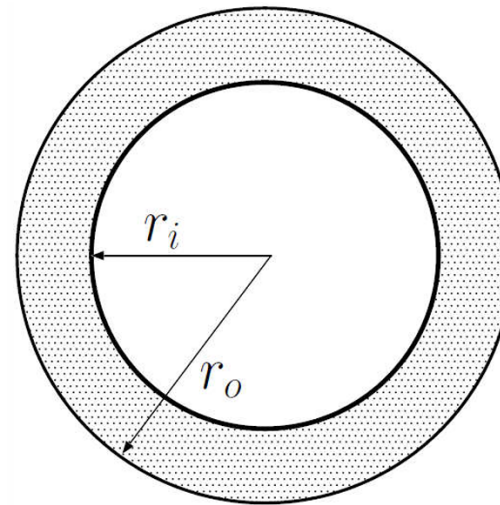
# Polar moment of inertia



Solid shaft

$$\begin{aligned} I_p &= \int_A \rho^2 dA \\ &= \int_0^{\frac{D}{2}} \rho^2 \cdot 2\pi \cdot \rho \cdot d\rho \\ &= \frac{\pi D^4}{32} = \frac{\pi}{2} r^4 \end{aligned}$$

Units of  $I_p$ :  $\text{length}^4$

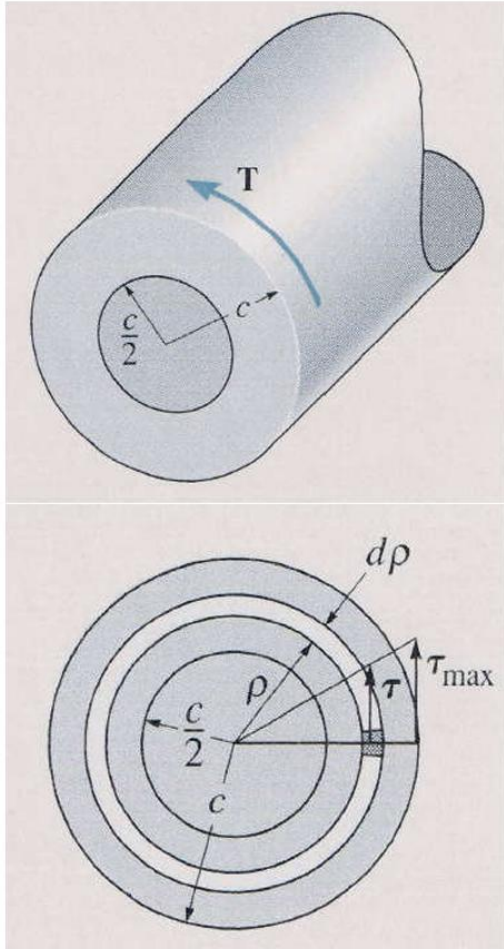


Tubular shaft

$$\begin{aligned} I_p &= \int_A \rho^2 dA \\ &= \int_{\frac{d}{2}}^{\frac{D}{2}} \rho^2 \cdot 2\pi \rho \cdot d\rho \\ &= \frac{\pi}{32} (D^4 - d^4) \end{aligned}$$



# Illustration of efficiency of tubular shaft



Consider  $T'$ , the contribution of  $\rho \in [\frac{c}{2}, c]$

$$dT' = \rho(\tau dA) = \rho \frac{\rho}{c} \tau_{\max} (2\pi \rho d\rho)$$

$$T' = \frac{2\pi \tau_{\max}}{c} \int_{c/2}^c \rho^3 d\rho = \frac{15\pi}{32} \tau_{\max} c^3$$

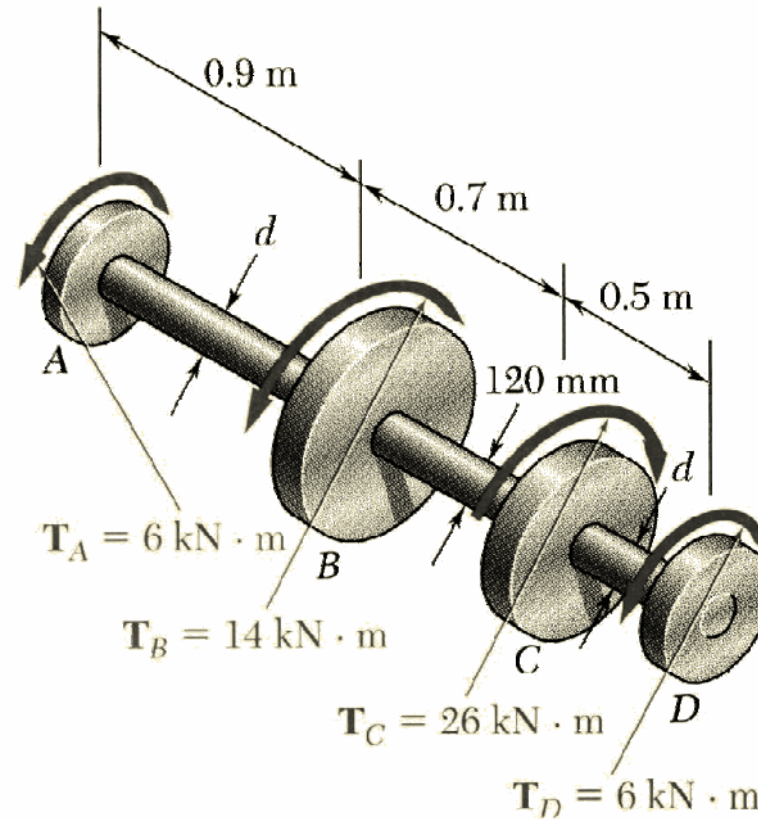
$$\tau_{\max} = \frac{Tc}{I_p} = \frac{Tc}{(\pi/2)c^4} = \frac{2T}{\pi c^3}$$

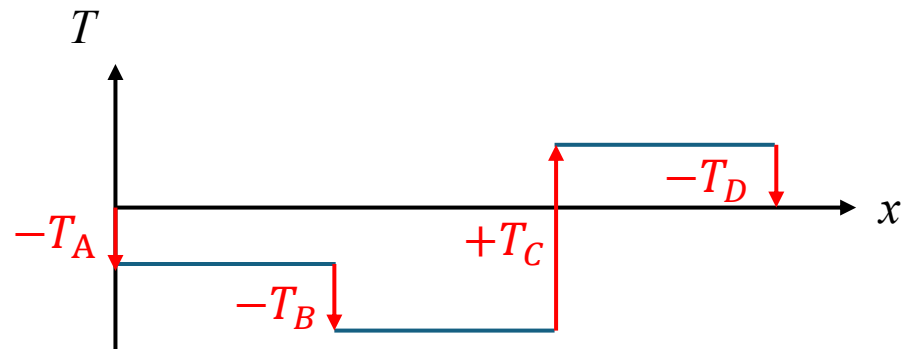
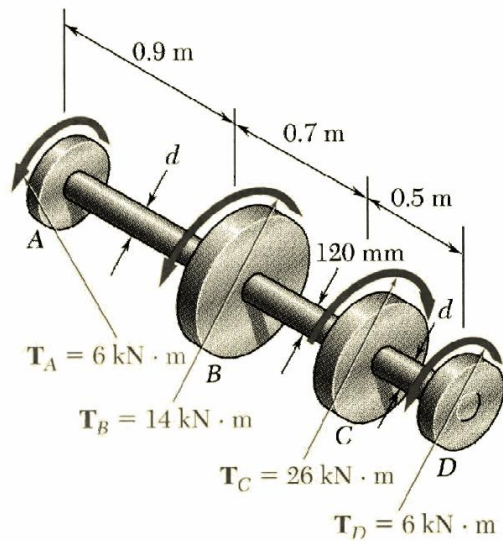
$$T' = \frac{15}{16} T \approx 0.94T$$

**Outer region** of the shaft bears most of the torsional load.

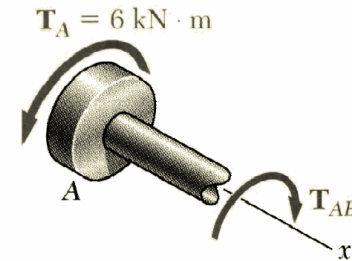
# Exercise

Shaft  $BC$  is hollow with inner and outer diameters of 90 mm and 120 mm, respectively. Shafts  $AB$  and  $CD$  are solid of diameter  $d$ . For the loading shown, determine (a) the minimum and maximum shearing stress in shaft  $BC$ , (b) the required diameter  $d$  of shafts  $AB$  and  $CD$  if the allowable shearing stress in these shafts is 65 MPa.



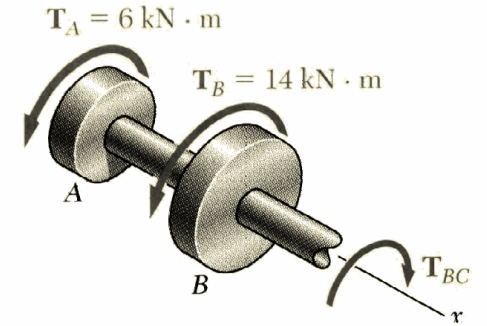


- Cut sections through shafts AB and BC and perform static equilibrium analysis to find torque loadings.



$$\sum M_x = 0 = (6 \text{ kN} \cdot \text{m}) - T_{AB}$$

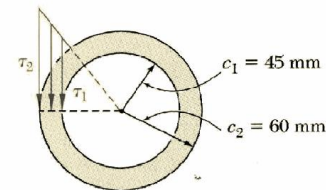
$$T_{AB} = 6 \text{ kN} \cdot \text{m} = T_{CD}$$



$$\sum M_x = 0 = (6 \text{ kN} \cdot \text{m}) + (14 \text{ kN} \cdot \text{m}) - T_{BC}$$

$$T_{BC} = 20 \text{ kN} \cdot \text{m}$$

- Apply elastic torsion formulas to find minimum and maximum stress on shaft BC.



$$I_P = \frac{\pi}{2} (c_2^4 - c_1^4) = \frac{\pi}{2} [(0.060)^4 - (0.045)^4]$$

$$= 13.92 \times 10^{-6} \text{ m}^4$$

$$\tau_{\max} = \tau_2 = \frac{T_{BC} c_2}{I_P} = \frac{(20 \text{ kN} \cdot \text{m})(0.060 \text{ m})}{13.92 \times 10^{-6} \text{ m}^4}$$

$$= 86.2 \text{ MPa}$$

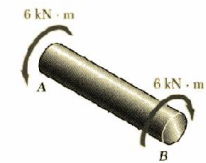
$$\frac{\tau_{\min}}{\tau_{\max}} = \frac{c_1}{c_2} \quad \frac{\tau_{\min}}{86.2 \text{ MPa}} = \frac{45 \text{ mm}}{60 \text{ mm}}$$

$$\tau_{\min} = 64.7 \text{ MPa}$$

$$\tau_{\max} = 86.2 \text{ MPa}$$

$$\tau_{\min} = 64.7 \text{ MPa}$$

- Given allowable shearing stress and applied torque, invert the elastic torsion formula to find the required diameter.

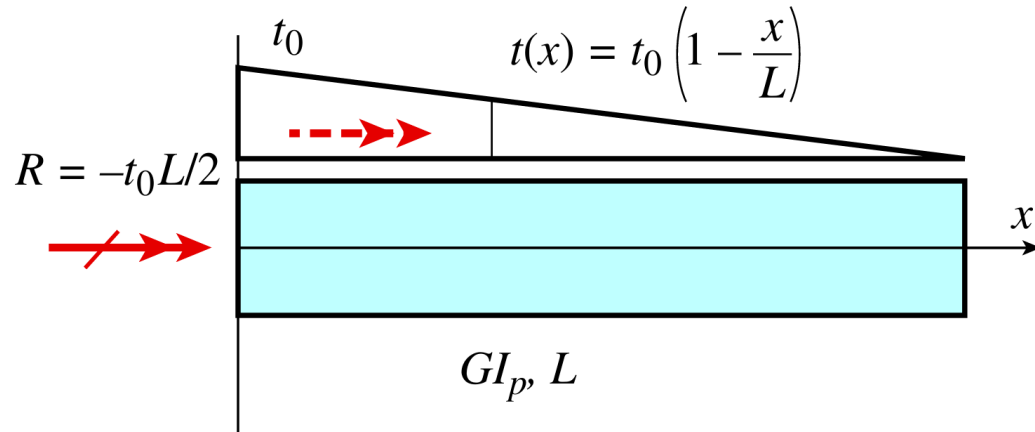


$$\tau_{\max} = \frac{Tc}{I_P} = \frac{Tc}{\frac{\pi}{2} c^4} \quad 65 \text{ MPa} = \frac{6 \text{ kN} \cdot \text{m}}{\frac{\pi}{2} c^3}$$

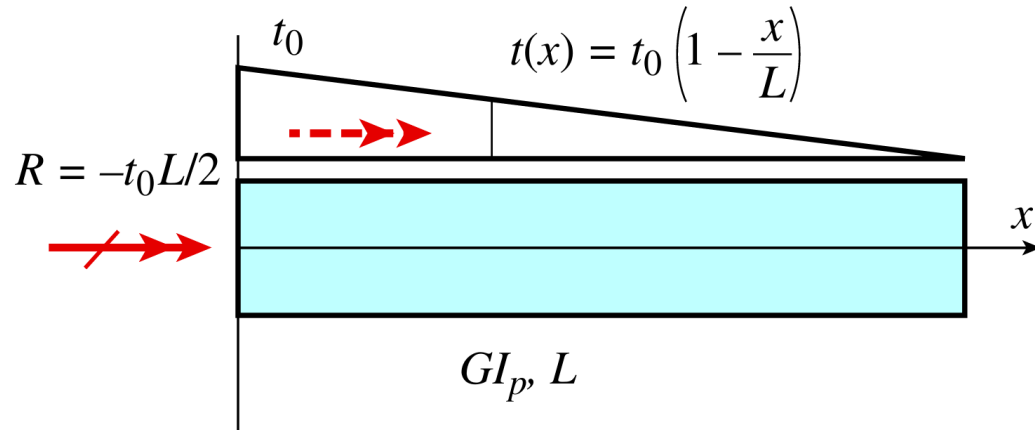
$$c = 38.9 \times 10^{-3} \text{ m}$$

$$d = 2c = 77.8 \text{ mm}$$

# Torsional displacement diagram (TDD)

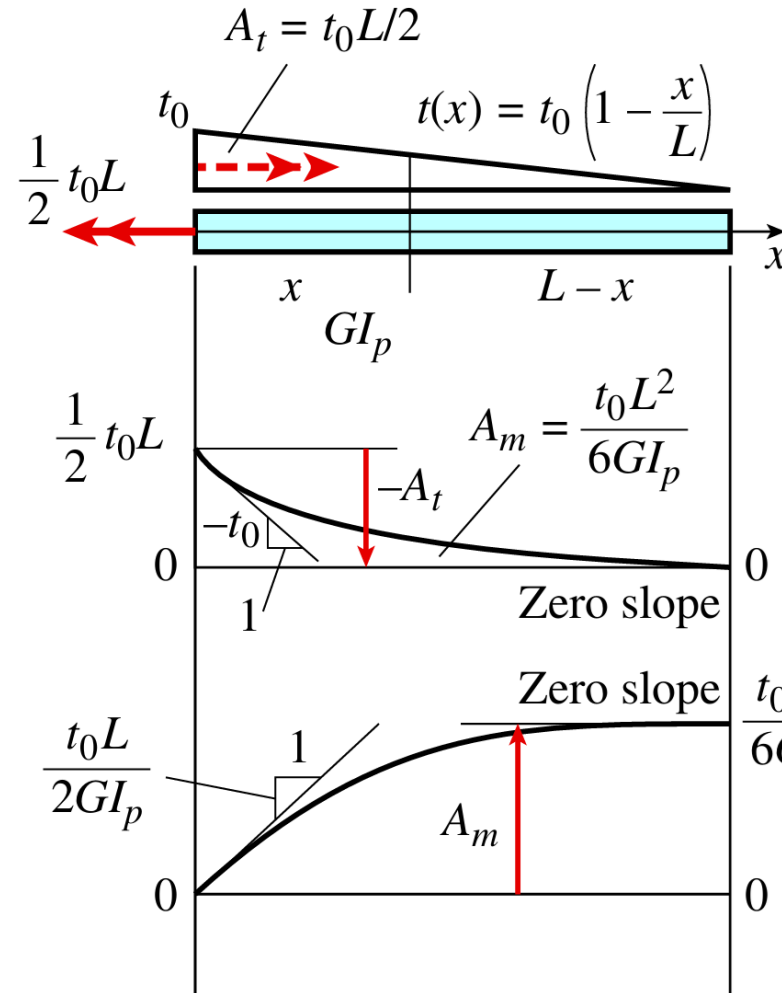


# Torsional displacement diagram (TDD)



$$\frac{dT}{dx} = t \Rightarrow T = \int_0^x t(\xi) d\xi$$

$$\theta = \frac{d\phi}{dx} = \frac{\tau}{G\rho} = \frac{T}{GI_p} \Rightarrow \phi = \int_0^x \frac{T(\xi)}{GI_p} d\xi$$

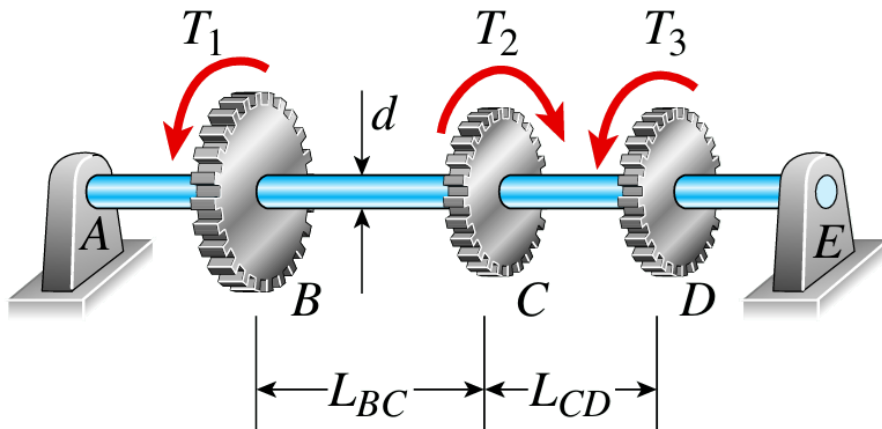


(b) Torsional load diagram

(c) Torsional moment diagram (TMD)

(d) Torsional displacement diagram (TDD)

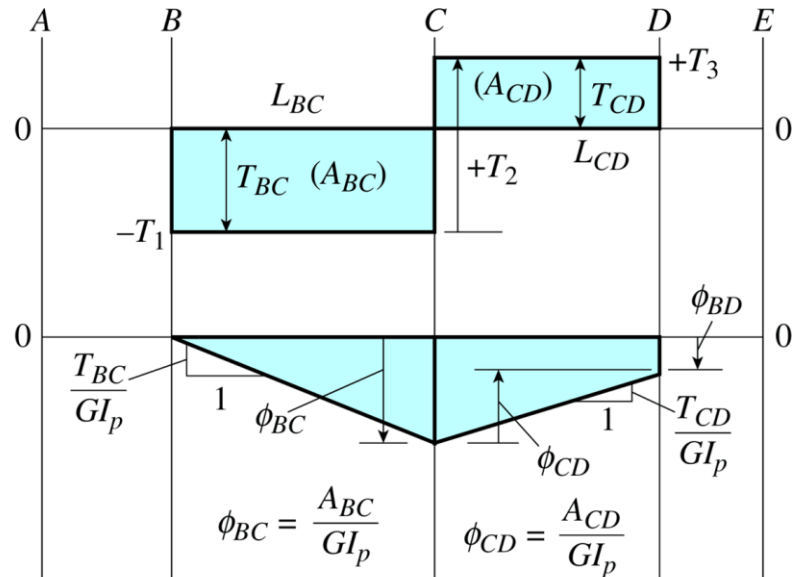
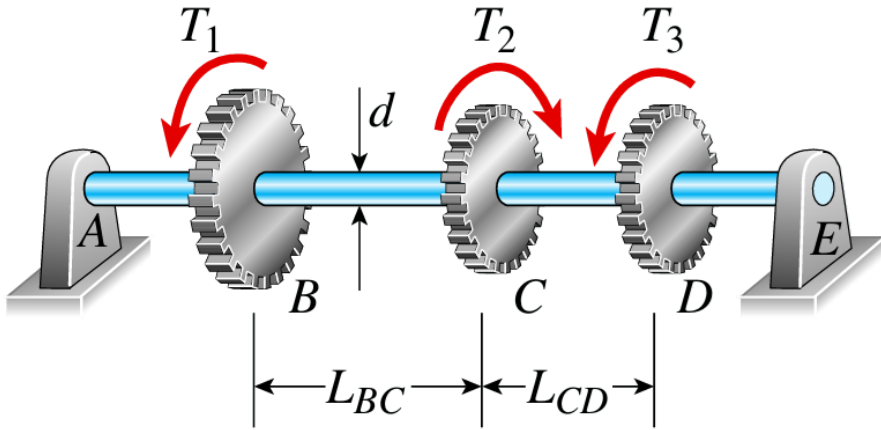
# Exercise



A solid steel shaft  $ABCDE$  (Fig. 3-18) having a diameter  $d = 30$  mm turns freely in bearings at points  $A$  and  $E$ . The shaft is driven by a gear at  $C$ , which applies a torque  $T_2 = 450$  N·m in the direction shown in the figure. Gears at  $B$  and  $D$  are driven by the shaft and have resisting torques  $T_1 = 275$  N·m and  $T_3 = 175$  N·m, respectively, acting in the opposite direction to the torque  $T_2$ . Segments  $BC$  and  $CD$  have lengths  $L_{BC} = 500$  mm and  $L_{CD} = 400$  mm, respectively, and the shear modulus is  $G = 80$  GPa.

Determine the maximum shear stress in each part of the shaft and the angle of twist between gears  $B$  and  $D$ .

# Exercise



$$\tau_{BC} = \frac{16T_{BC}}{\pi d^3} = \frac{16(275 \text{ N} \cdot \text{m})}{\pi(30 \text{ mm})^3} = 51.9 \text{ MPa}$$

$$\tau_{CD} = \frac{16T_{CD}}{\pi d^3} = \frac{16(175 \text{ N} \cdot \text{m})}{\pi(30 \text{ mm})^3} = 33.0 \text{ MPa}$$

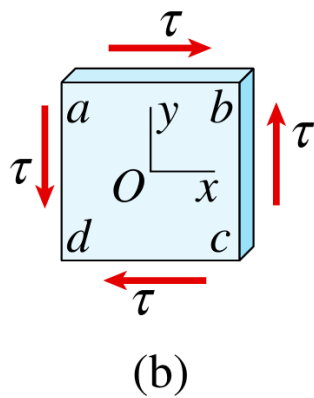
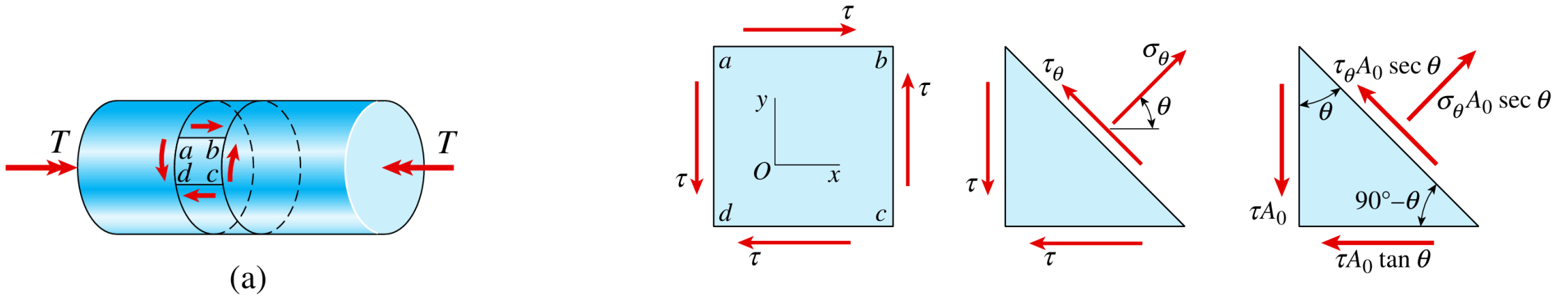
$$I_p = \frac{\pi d^4}{32} = \frac{\pi(30 \text{ mm})^4}{32} = 79,520 \text{ mm}^4$$

$$\phi_{BC} = \frac{T_{BC}L_{BC}}{GI_p} = \frac{(-275 \text{ N} \cdot \text{m})(500 \text{ mm})}{(80 \text{ GPa})(79,520 \text{ mm}^4)} = -0.0216 \text{ rad}$$

$$\phi_{CD} = \frac{T_{CD}L_{CD}}{GI_p} = \frac{(175 \text{ N} \cdot \text{m})(400 \text{ mm})}{(80 \text{ GPa})(79,520 \text{ mm}^4)} = 0.0110 \text{ rad}$$

$$\phi_{BD} = \phi_{BC} + \phi_{CD} = -0.0216 + 0.0110 = -0.0106 \text{ rad} = -0.61^\circ$$

# Stress on inclined surfaces

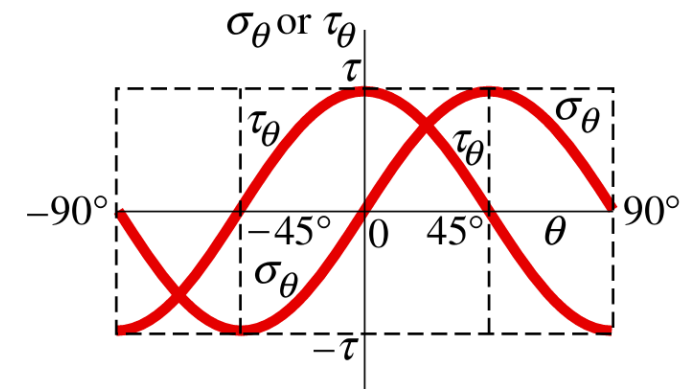


$$\begin{aligned}\sigma_{\theta} A_0 \sec \theta &= \tau A_0 \sin \theta + \tau A_0 \tan \theta \cos \theta \\ \tau_{\theta} A_0 \sec \theta &= \tau A_0 \cos \theta - \tau A_0 \tan \theta \sin \theta\end{aligned}$$



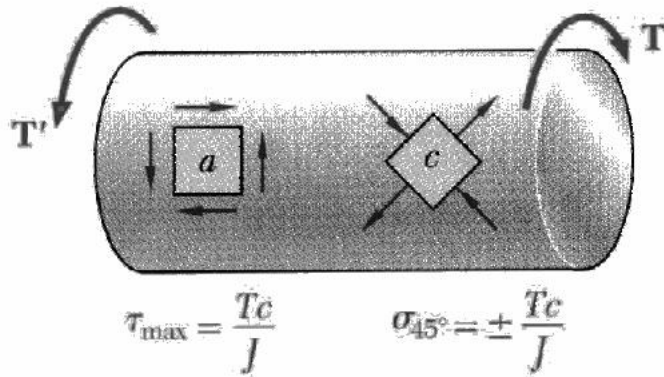
$$\begin{aligned}\sigma_{\theta} &= 2\tau \sin \theta \cos \theta \\ \tau_{\theta} &= \tau(\cos^2 \theta - \sin^2 \theta)\end{aligned}$$

$$\sigma_{\theta} = \tau \sin 2\theta \quad \tau_{\theta} = \tau \cos 2\theta$$

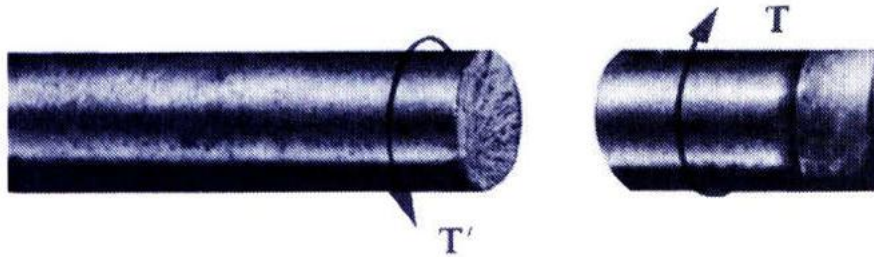




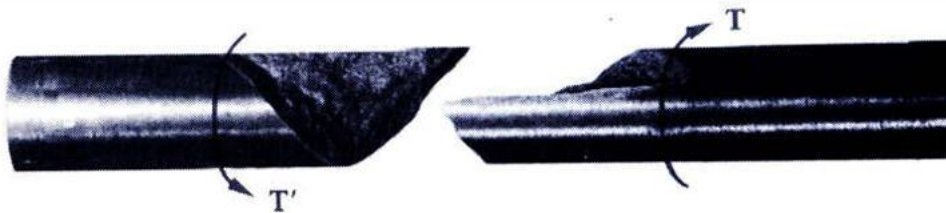
# Torsional failure modes



Ductile materials generally fail in shear. Brittle materials are weaker in tension than shear.



Ductile materials fail in shear (90° fracture)



Brittle materials are weaker in tension (45° fracture)